Dutpal Ananya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.10
Sri Chaitanya Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	98.20%
Sri Chaitanya School	Class X (Board of Secondary Education, Telangana)	2021	100.00%

WORK EXPERIENCE

AI-ML Intern | Country Delight

- Built an **EfficientNet**-based **image classifier** that sorts incoming coconut complaint images into **6 defect categories**, helping the **customer care** team triage faster and resolve issues on the company's **2nd best-selling product**.
- Engineered an **LLM prompt** that categorizes customer chat conversations and shows which of **8 chatbot APIs** should be split or newly created, informing chatbot design while **reducing token costs**.
- Developed a **retail audit pipeline** combining an **EfficientNet** validity filter to reject unusable shelf images with a **YOLO** model that counts products, and deployed a **Hugging Face UI** enabling bulk uploads from **S3**.
- Trained a **win-back model** on **21 user attributes** to score churned users by their likelihood of returning, giving marketing a **ranked list** to prioritize instead of contacting every lapsed user.
- Targeted the top-scored users in **re-engagement campaigns**, lifting **conversion from under 1% to 6%** (about **6x**) while **reducing outreach spend** by focusing on high-intent users.
- Prototyped a **tomato ripeness detector** using **EfficientNet** and **YOLO**, and evaluated **business fit** and **production feasibility**, learning how cost, data availability, and ROI determine whether a model moves beyond the prototype stage.

Data Science Intern | Bheri *(Prof. Mohan Raghavan, IIT Hyderabad)*

- Processed **10,000+ Sanskrit verses** from **100+ GitHub-hosted texts** into structured **datasets** for downstream **AI applications**.
- Built **vector-embedding pipelines** to convert Sanskrit verses into embeddings, enabling **semantic search** across dataset.
- Applied **semantic similarity**, **clustering** and **retrieval** techniques to group related verses and surface relevant passages.

PROJECTS

Neuro-Symbolic Sequence Solver | Link to Project

- Built a **hybrid solver** (**94.59% accuracy**) that plays Hangman on unseen words: it filters a **200K+ word list**, uses a **PyTorch BiLSTM** for out-of-vocabulary words, and falls back to letter frequencies.
- Sped up word filtering with **26-bit integer bitmasks**, so each letter check is a single **O(1)** bitwise operation.

City Traffic Network Simulator | **Link to Project**

- Simulated **up to 1,400 cars** in a **144-intersection, 255-road city** where roads slow down as they fill up; wrote **Dijkstra** and **A*** from scratch (A* explores **67% fewer nodes**, same route).
- Compared cars that **never change route** with cars that **re-check at every intersection**: re-routing cut average trip time by **50%** at 1000 vehicles; live **browser dashboard** over WebSockets, **19 tests and CI**.

minikv: In-Memory Key-Value Store (like Redis) | **Link to Project**

- Built a **fast in-memory database in C++**: when full, it removes the **least recently used** key in **O(1)** (**hash map + doubly linked list**), and keys **expire automatically** using a **min-heap**.
- Added **nested transactions** (BEGIN / ROLLBACK / COMMIT) with undo logs, an interactive shell and a **Python REST API**; verified with **44 tests**, including random tests against simple reference models.

SKILLS

Programming: Python, C++, C, SQL, Data Structures & Algorithms

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Computer Vision (YOLO, EfficientNet)

Tools: Git, GitHub, BitBucket, Jupyter, Google Colab, VS Code

POSITION OF RESPONSIBILITY

Head of Finance & Corporate Partnerships | Finance and Consulting Club, IIT Hyderabad

- Promoted from Coordinator to **Head and** Led a team of **10 coordinators** and secured **10 sponsors** for flagship event **HORIZON.**
- Raised **Rs.10+ lakhs** in monetary partnerships & secured multiple non-monetary partnerships, helping scale the event **10x**.
- Ran the on-campus **Ankur Warikoo talk** end to end (**800+ attendees**), handling outreach, logistics and sponsor deliverables.
- Coordinated with **Design, Operations and Marketing teams** to ensure every partner company's deliverables were met.

ACHIEVEMENTS AND ADDITIONAL INFORMATION

- Ranked **1st** on the leaderboard at the **Meltwater Hackathon**, IIT Hyderabad.
- Trained **Kuchipudi dancer** with over **6+ years of experience and** with **50+ stage performances**
- Represented **IIT Hyderabad** in **Dance** at Inter-IIT Cultural Meet 7.0, securing **10th position among 23 IITs**.
- Represented **IIT Hyderabad** in **Culinary** at Inter-IIT Cultural Meet 8.0, placing **5th position among 23 IITs**.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Applications, Algorithms and Data Structures Lab, Artificial Intelligence , Foundations of Natural Intelligence .

Mathematics: Calculus, Linear Algebra, Differential Equations, Transform Techniques, Introduction to Probability, Introduction to Statistics, Bayesian Inference, Statistical Inference .